Deletion, Arrows, Windows and Composition: a Classification of Structural Patterns in 10,404 Certified Polytope Packings

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Abstract

The Polyhedra-in-Polyhedra catalogue holds 10,404 certified packings of n congruent convex polytopes in a scaled container, across 140 problem families in dimensions 3–6. Records are found by many searches and contributors, but they are far from structureless. We classify the recurring patterns into nine classes: deletion and plateaus, subset relaxation ("arrows"), insertion and completion, composition and recursion, lattice windows, layers and orientation classes, symmetry, algebraic contact scales, and extremal (provably optimal) configurations. For each class we give a precise mechanism, prove it in Lean 4 with Mathlib where it is a theorem about packings, and measure its prevalence over the whole catalogue. The general theorems are deletion, scale monotonicity, composition $s(nm) \le s(n)\, s(m)$, separator certificates and Voronoi windows. Exact instances are formalized at their exact scales: every finite FCC window of rhombic dodecahedra (19, 44 and 93 pieces at scales 3, 4 and 5), the 56-piece Cartesian shell of truncated cubes at scale 4, and the triakis-tetrahedron recursion 4^k pieces at $(5/3)^k$. About a third of all records (3,397) lie on 861 plateaus of constant scale. 889 3D records are lattice windows, and 776 have nontrivial configuration symmetry. PingYou Ltd recently described arrows: select bodies of a larger packing, then hold orientations fixed and move centres (an exact linear program), and finally release all coordinates. Applied to catalogue records alone, the same procedure reproduces six of their new square-antiprism bounds ($n = 3$ and 5–9) digit for digit, through subsets, a kick, and a chain of insertions.

1 Introduction

A *finite container packing* asks for the smallest scale $s(n)$ of a container C holding n congruent copies of a body K with disjoint interiors. The Polyhedra-in-Polyhedra catalogue [1] records, for 140 pairs (K, C) of Platonic, Archimedean and Catalan solids, prisms, antiprisms and regular polytopes in dimensions 4–6, the best certified packing found for each n up to 100: 8,514 records in 3D and 1,404, 279 and 207 in dimensions 4, 5 and 6. Every record carries an exact certificate of containment and pairwise separation. The packings come from soft-to-rigid continuation [4, 2], basin hopping, lattice cuts, composition, published constructions [5] and many community contributions, prominently those of PingYou Ltd, `konoha-ily2` and PolyPack.

Looking at the catalogue as data, the records are far from independent optima. They come in plateaus where $s(n)$ does not change. They are finite windows of lattice tilings, or compositions of smaller packings. They inherit large rigid blocks from neighbouring records. They show layered and symmetric arrangements, and their scales satisfy short algebraic equations. PingYou Ltd recently made two such observations explicit: an illustrated account of sixteen exact constructions and their number patterns [7], and "arrows" that recover smaller square-antiprism packings from a larger one [8]. This paper turns these observations into a classification with three layers:

Class	Mechanism	Formal statement (Lean)	Prevalence
P1 Deletion, plateaus	$m \le n$ pieces of an n-packing are an m-packing	`Packing.restrict`	861 plateaus, 3,397 records; 1,438 records built by deletion
P2 Arrows	subset, then F (fixed orientations, LP), then P (free); insertions, kicks	`Packing.restrict`, `Packing.of_separators`	6 of PingYou's square-antiprism bounds reproduced from records
P3 Insertion, completion	$(n-1)$-packing plus a piece in a hole	(constructive only)	1,974 records built by insertion
P4 Composition	$s_C(Q,n)\, s_Q(P,m)$ bounds $s_C(P,nm)$	`Packing.compose`, `Packing.pow`	15 exact composites, 63 nested records
P5 Lattice windows	Voronoi cells of distinct points	`voronoi_packing`, `lattice_window`	889 windows (407 at rational scale)
P6 Layers, orientation classes	a few layers, few orientations	(descriptive)	1,241 single-orientation records
P7 Symmetry	invariance under container symmetries	(descriptive)	776 symmetric 3D records
P8 Algebraic scales	contact equations close the scale	(exact certificates)	1,624 closed forms
P9 Extremal	lower bound meets construction	centre lemmas, volume	proven optimal records (Section 11)

Table 1: The nine pattern classes. Prevalence counts are over all 10,404 records, or over the 8,514 3D records where a geometric feature is needed.

1. **Mechanisms.** Each pattern class is a construction rule. Where the rule is a statement about packings, we prove it once in Lean 4 [10] for all bodies, containers and dimensions (Section 14).
2. **Exact instances.** Catalogue records that realize a mechanism exactly are formalized at their exact scales, rather than the clearance-padded scales of the certified records.
3. **Prevalence.** Each class is measured over the whole catalogue by reproducible scripts (`scripts/patterns/`), and every number in this paper is generated from their output.

Table 1 summarizes the nine classes. The remaining sections treat them in turn, then classify the families (Section 13) and state the limits of the claims (Section 15).

2 Setting

Let E be a finite-dimensional real inner product space, $K, C \subseteq E$ and $s \ge 0$. A *packing* of n copies of K in sC is a family of affine isometries $g_1, \dots, g_n$ of E with $g_i(K) \subseteq sC$ and $\operatorname{int} g_i(K) \cap \operatorname{int} g_j(K) = \emptyset$ for $i \ne j$. This is the structure `Packing K C s n` of the Lean development (`lean-optimality/PackingOptimality/Basic.lean`). Reflections are allowed, so lower bounds proved for it hold for the catalogue's rotation-only packings. The catalogue measures s as the ratio of container to piece edge (or linear size for Catalan solids). A record r stores $s_{\text{full}}(r)$, carrying a clearance of at least 10^{-7} between pieces and walls, and an exact certificate. The touching limit s_{tight} is what the patterns below are about.

3 Deletion, plateaus and frozen subsets (P1)

The simplest mechanism is deletion: any $m \le n$ pieces of an n-packing form an m-packing in the same container, so $s(m) \le s(n)$ (`Packing.restrict`). The catalogue enforces this monotonicity: when a search finds $s(n+1) < s(n)$, the $(n+1)$-packing minus one piece replaces the n-record. 1,438 records arose this way, and 44 more from an explicit deletion move.

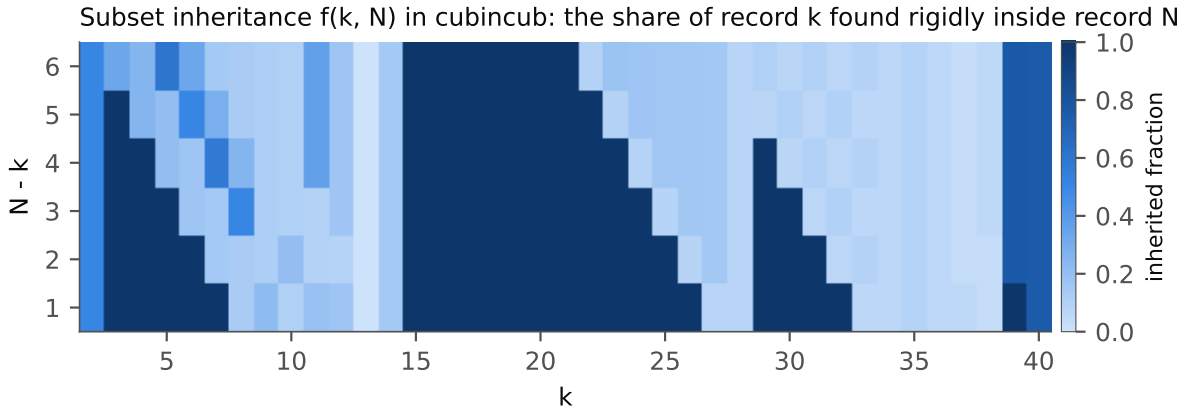


Figure 1: Subset inheritance in cubes in a cube. Dark cells: record k is a rigid subset of record N. The plateaus $s = 2$ ($n \leq 8$), $s = 3$ ($15 \leq n \leq 27$) and $s = 4$ ($39 \leq n \leq 64$) appear as solid blocks.

Plateaus. A plateau is a maximal run $n_0 < \cdots < n_1$ with $s(n_0) = \cdots = s(n_1)$. The catalogue has 861 plateaus containing 3,397 records. The longest are $s = 2$ for $2 \leq n \leq 32$ five-cubes and $2 \leq n \leq 24$ six-cubes (Hosono's centre lemma [6]). Others are $s = 4$ for $39 \leq n \leq 64$ cubes and $41 \leq n \leq 64$ triangular prisms (grid tilings), $s = 4$ for $37 \leq n \leq 56$ truncated cubes (the Cartesian shell, Section 7) and $s = 5$ for $75 \leq n \leq 93$ rhombic dodecahedra. A plateau ends at a *window number*, the largest count a rigid construction holds at that scale. Most plateaus are deletions of a single lattice window or tiling.

Frozen subsets. Deletion predicts a sharper relation between records. For records $k < N$ of one family, let $f(k, N)$ be the largest fraction of the k pieces that can be matched one-to-one with pieces of the N-packing after a single container symmetry and translation, with matching centres and identical placed vertex sets (orientation included, up to the piece's own symmetry; tolerance 0.02 edges; `scripts/patterns/inherit.py`). Then $f = 1$ means record k is literally k pieces of record N, moved rigidly. Over 20,124 pairs with $N - k \leq 6$ in 86 families, 1,384 pairs are frozen ($f = 1$), 601 of them consecutive ($N = k + 1$). Figure 1 shows the cube family. Frozen blocks appear exactly on the plateaus and on the k^3 grids. Between them, consecutive records share little rigidly: each was re-optimized after the deletion. Over all families, frozen pairs thin out with the gap: 601 of the 3,354 consecutive pairs are frozen, and fewer at each larger gap.

4 Arrows: subset relaxation and kicks (P2)

PingYou Ltd observed that a larger packing contains good smaller ones that are *not* rigid subsets: selecting six or eight square antiprisms of their 19-piece packing and relaxing them recovers their 6- and 8-piece bounds [8]. They specified the relaxation as two steps, with the unit bodies fixed throughout:

F hold every orientation fixed and minimize the scale over the centres, re-selecting pair separation planes;

P minimize the scale with centres and rotations free, by a sequential linear program.

Arrows combine these with discrete moves: deletion (keep a subset), insertion (add a body at an affine guess or at the hub, the mean of the centres), and kicks (rotate one body about its own axis, or a group of bodies about an axis through their centroid, by $\pm 6°$ to $\pm 16°$), each followed by F then P.

n	catalogue record	arrow from catalogue records	scale reached	PingYou Ltd [8]
3	1.973017314	subset $\{4, 6, 12\}$ of $N = 15$, F, P	1.9726490831	1.972649083082090
5	2.209265853	subset of $N = 9$, F, P	2.1785815015	2.178581501510501
6	2.292893610	subset of $N = 9$, F, P; spin 16°, F, P	2.2459217996	2.245921799598007
7	2.373563494	insert into $n = 6$ (affine guess); four spins, F, P after each	2.3629758095	2.362975810
8	2.402020630	insert into $n = 7$; four spins	2.3849425743	2.384942574336931
9	2.542195713	hub insertion into $n = 8$, F, P	2.4890416764	$1 + 2/(11 - 4\delta)$
10	2.590076603	spin 6° of the record, F, P	2.5900761895	2.590076189471171
11	2.743255848	kicks of the record, F, P after each	2.7406579330	2.652646434968480
12	2.807780046	kicks of the record	2.8008665059	2.764993777408391
13	2.907150931	kicks of the record	2.8905573692	2.842219125707920
14	2.970067187	kicks of the record	2.9543988095	2.945858498562152

Table 2: Arrows on square antiprisms in a square antiprism, from catalogue records only. "Scale reached" is the touching limit after F and P. The certified records add 10^{-7} clearance.

F is an exact linear program. With orientations R_i fixed and one unit axis u_{ij} chosen for each pair, write $h_i(u) = \max_{x \in K} \langle u, R_i x \rangle$. Every constraint is then linear in the centres c_i and the scale s:

$$\langle N_f, c_i \rangle + h_i(N_f) \le \beta_f \, s \quad \text{(walls)}, \qquad \langle u_{ij}, c_i \rangle + h_i(u_{ij}) \le \langle u_{ij}, c_j \rangle - h_j(-u_{ij}) \quad \text{(pairs)}, \quad (1)$$

where $\{x : \langle N_f, x \rangle \le \beta_f\}$ are the container's facets. Any feasible point is a packing, since each pair is split by a plane: this is `Packing.of_separators`, proved for arbitrary bodies. Its optimum is therefore a certified packing for that choice of axes. F iterates by re-selecting each pair's best separating axis at the current point; the point stays feasible, so the scale never increases (`scripts/patterns/fp.py`).

Reproduction from catalogue records. We ran arrows on the square-antiprism family ap4 using only catalogue records as sources. The searches tried subsets of records $N \le 20$ chosen as caps along container directions, nearest clusters and random connected clusters, then insertion and kicks from the improved packings (`scripts/patterns/arrows_search.py`, `arrows_kick.py`). Table 2 lists the results. For $n = 3$ and 5–9 the arrows reach PingYou's new bounds to every digit they published, from different sources: their 19-piece packing is not in the catalogue. For $n = 9$ their value is the closed form $1 + 2/(11 - 4\delta)$, $\delta = 1 + \sqrt{2}$. For $n = 6$ the subset alone stops at 2.2500. A single kick, a 16° spin of one body about its own axis followed by F and P, reaches their value, which they had obtained by selecting six bodies of their 19-piece packing. For $n = 10$ a single 6° spin of the record reproduces their tighter certificate. For $n = 11$–14 kicks of the catalogue record improve it, but stay above PingYou's values. Those came from different constructions: relative square-face spins with receiver reflow, joint pair reinsertion, and ordered overlapping patches. The chain $6 \to 7 \to 8 \to 9$ follows their insertion arrows $N6 \to N7$ and $N8 \to N9$. Coincident values from different starting points indicate the same rigid contact configuration, reached along different arrows. The reproduced packings are now catalogue records, with 10^{-7} clearance and exact certificates, crediting PingYou Ltd as first reporter; the witnesses are reproductions by their procedure. The weaker improvements for $n \ge 11$ are recorded with PingYou's stronger announced values noted.

5 Insertion and completion (P3)

The converse of deletion adds a piece to an $(n-1)$-packing, in its largest hole, and re-sharpens (1,974 records). A rigid version is *completion*: a construction with empty corners or gaps receives additional pieces at the same or slightly larger scale. PingYou's 61 truncated cubes [7] complete the 56-piece Cartesian shell (Section 7) with five corner pieces. The 40 wall and contact inequalities of their contact system sum to $5(2 + \sqrt{2})(S - 1) \ge 60$, so $S \ge 13 - 6\sqrt{2}$, and the

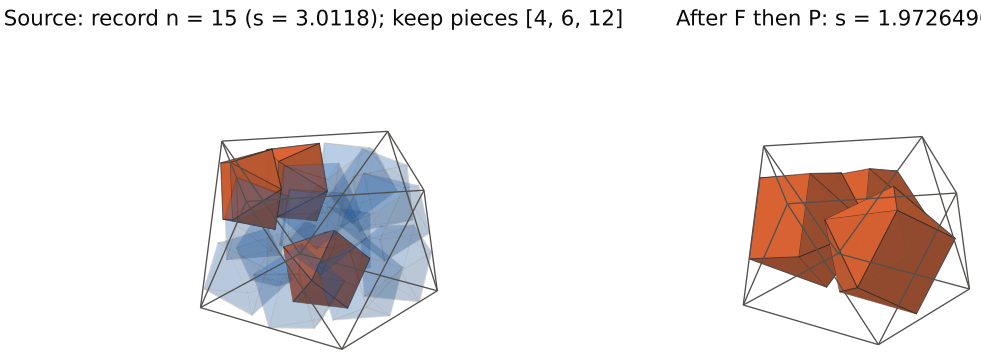


Figure 2: An arrow: three square antiprisms of the 15-piece record, then F and P, reach $S_3 = 1.9726490831$.

construction attains it. That bound holds for the declared contact system, not for all packings. Insertion arrows (PingYou's $N6 \to N7$ and $N8 \to N9$, the latter with a hub body of orientation $\mathrm{diag}(1,-1,-1)$) are completion followed by F and P.

6 Composition and recursion (P4)

Scales are ratios, so they multiply.

Theorem 1 (`Packing.compose`)**.** *If n copies of Q pack in $s\,C$ and m copies of P pack in $t\,Q$ with $t \neq 0$, then nm copies of P pack in $(st)\,C$. In particular, for $K = P = Q = C$, $s(nm) \leq s(n)\,s(m)$ (`Packing.selfCompose`) and $s(n^k) \leq s(n)^k$ (`Packing.pow`).*

Proof sketch. Conjugating g by the dilation $x \mapsto tx$ gives $\hat{g} = x \mapsto t\,g(0) + \vec{g}(x)$ with $\hat{g}(tA) = t\,g(A)$. The pieces $\hat{g}_i(h_j(P)) \subseteq t\,g_i(Q) \subseteq (st)\,C$ are disjoint within one g_i because $\hat{g}_i$ is an isometry. Across $i \neq i'$ they are disjoint because the interiors of $t\,g_i(Q)$ and $t\,g_{i'}(Q)$ are. □

Nested composition is how 63 records were built (`scripts/compose3d.py`, `compose_nd.py`; the whole-body inclusion idea is PingYou's contribution [9]). In the self-similar families, 15 records equal a product $s(n)s(m)$ exactly. They include the cube and prism tilings ($64 = 8 \cdot 8$ at scale 4) and the rhombic-dodecahedron record $36 = 6 \cdot 6$ at 4. The cleanest is the triakis tetrahedron, where PingYou recognized the 16- and 64-piece records as the 4-piece packing squared and cubed [7].

Theorem 2 (`triakis_recursion`)**.** *For every $k \geq 0$, 4^k triakis tetrahedra pack in a triakis tetrahedron of scale $(5/3)^k$.*

Proof. Write the catalogue's triakis tetrahedron as $\{x : \langle m, x\rangle \leq 9\sqrt{2}/4\}$ over its twelve integer face normals m (of type $(1,3,1)$). Place piece i at $R_iK + (\sqrt{2}/2)\sigma_i$ with $\sigma_i \in \{(-1,1,1),(-1,-1,-1),(1,1,-1),(1,-1,1)\}$ and rotations R_i with entries $\pm\frac{1}{3}, \pm\frac{2}{3}$. Containment in $\frac{5}{3}K$ is a linear consequence of the face inequalities. The six pairs touch along the planes $x_a \pm x_b = 0$. This gives the 4-piece packing at $5/3$ (`Triakis.ktt4`); Theorem 1 gives the powers. □

7 Lattice windows and tilings (P5)

Theorem 3 (`voronoi_disjoint`, `voronoi_packing`)**.** *If A lies in the Voronoi half-space of p against q and B in that of q against p ($p \neq q$), then A and B have disjoint interiors. Hence*

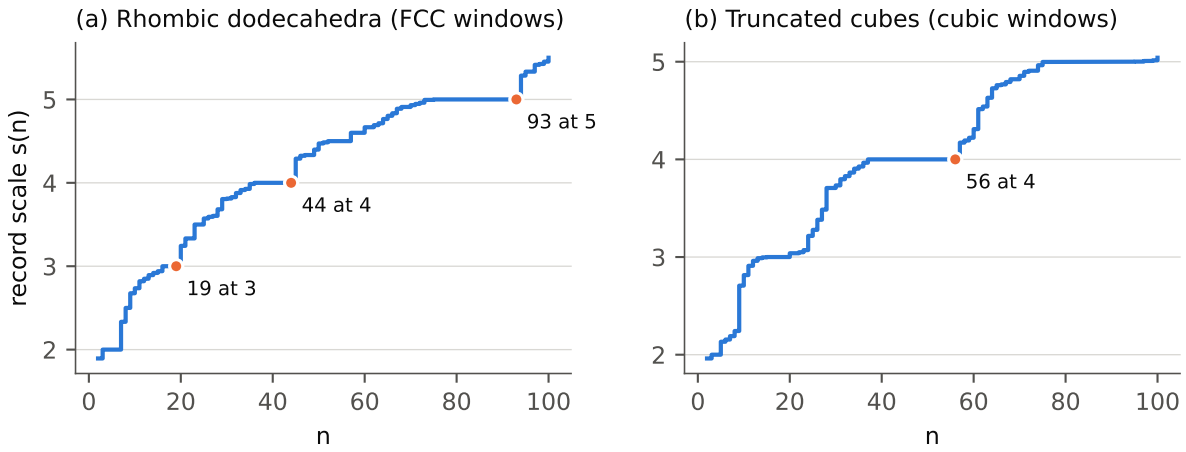


Figure 3: Record scales against n. Plateaus end at lattice window numbers: 19, 44 and 93 FCC windows of rhombic dodecahedra at scales 3, 4 and 5, and the 56-piece Cartesian shell of truncated cubes at scale 4 (all proved in Lean at the exact scales).

copies $g_i(K)$ lying in the Voronoi cells of distinct centres c_i, inside sC, form a packing. For a lattice Λ whose Voronoi cell contains K, any finite set of translates of K by points of Λ that fit in sC is a packing (`lattice_window`).

The proof separates A and B by the bisector $\langle q-p, x\rangle = (\|q\|^2 - \|p\|^2)/2$. The theorem turns every lattice window into a finite containment check, and it covers multi-orientation windows whose pieces lie in the Voronoi cells of their centres. We formalize two families completely (`Windows.lean`):

- **Rhombic dodecahedra.** $R = \{|x_i| + |x_j| \le 1\}$ is the Voronoi cell of D_3, the FCC lattice. For $x \in R$ and $v \in D_3$ one has $2\langle x, v\rangle \le \|v\|^2$. The support function of R is $\max(\|v\|_\infty, \|v\|_1/2)$, and both terms are at most $\|v\|^2/2$ on $D_3 \setminus 0$ (`rd_voronoi`). Every window $\{w \in D_3 : |2w_i + e_i| + |2w_j + e_j| \le M\}$ packs in $(M/2+1)R$ (`rdo_window`). The catalogue's records of 19, 44 and 93 pieces at scales 3, 4 and 5 are such windows: the FCC shell $1 + 12 + 6$ and the palindromic window $2 + 8 + 12 + 12 + 8 + 2$ of PingYou [7], and the 93-piece record. They are proved at the exact scales, and by deletion every smaller count at the same scale.
- **Truncated cubes.** The unit-edge truncated cube T lies in the Voronoi cell $[-a, a]^3$ of $2a\mathbb{Z}^3$, with $a = (1+\sqrt{2})/2$. The $4 \times 4 \times 4$ grid minus its eight corners packs 56 copies in $4T$ (`tcu56`): PingYou's Cartesian shell $56 = 4^3 - 2^3$, the end of the plateau $37 \le n \le 56$.

Over the 3D catalogue, 889 records are lattice windows: every orientation class with at least four pieces sits on a lattice, with at most four classes. 685 of them have a single orientation, and 407 sit at a rational scale. Figure 3 shows how the window numbers terminate the plateaus. Two-orientation windows include PingYou's truncated-tetrahedron windows: $35 = 19 + 16$ at $11/3$ and $71 = 40 + 31$ at $13/3$, the two counts being coefficients of $(1 + x + x^2)^4$ and $(1 + x + x^2 + x^3)^4$ [7].

8 Layers and orientation classes (P6)

Many records are stacks of a few layers, with only a few orientations of the pieces up to the piece's own symmetry. Over the 3D catalogue, 1,241 records use a single orientation and 507 use exactly two (Figure 4b). Layer inventories along a container direction give the number patterns

angle	$\cos\theta$	class pairs	main families
93.841°	$(\sqrt{3}-2)/4$	662	triangular prisms (heights 1, 1.5, 2, 3) in a cube
37.399°	$(4\sqrt{5}-1)/10$	167	dodecahedra in a dodecahedron, icosahedron, octahedron
104.478°	$-1/4$	112	triangular prisms in a cube
15.631°	–	98	triangular and hexagonal prisms in a cube (twist about the prism axis)
95.480°	$(\sqrt{5}-3)/8$	48	pentagonal prisms in a cube
62.799°	$(2\sqrt{2}-1)/4$	46	cubes in a cube
38.942°	$7/9$	36	octahedra in a cube

Table 3: Recurring non-rational relative orientations between orientation classes. The cosines are matched to 3×10^{-6} and are conjectural; one cluster is not identified at this precision.

PingYou highlights. Examples are $29 = 9 + 11 + 9$ and $30 = 9 + 12 + 9$ cubes in a dodecahedron, whose shared row phase forces the scales $5/\varphi^2$ and $23 - 13\varphi$. Others are $32 = 8 + 6 \cdot 4$ cubes in an icosahedron at $8/\varphi^2$, $44 = 6 + 6 + 32$ octagonal prisms at $4 + 2\sqrt{2}$, and $20 = 2 \times 2 \times 5$ dodecagonal prisms at $15/2$ [7]. The extraction is in `scripts/patterns/features.py`. It groups centres into layers along each container facet normal, and pieces into orientation classes when $R_i^{-1} R_j$ is a proper symmetry of the piece.

Relative orientations. Between two orientation classes a, b, the relative rotation $R_a^{-1} R_b$ is defined modulo the piece's proper symmetries. We take the smallest angle over that coset (`scripts/patterns/rotations.py`). Over the 1,501 3D records with two to eight classes, 10,444 class pairs give 3,303 rational angles, that is, p/q of a full turn with $q \leq 24$. In 626 records every relative rotation is rational. The rational angles are the pieces' own symmetry angles: 90° (1/4 turn), 60°, 30°, 36°, 45°, 18°, 15° and 22.5°, in decreasing frequency. The remaining angles are not random either. They cluster at a few algebraic values, each tied to a family (Table 3). Those values compose a symmetry rotation of the piece with a symmetry rotation of the container. Examples are $\arccos(-1/4)$ and $\arccos((\sqrt{3}-2)/4)$ for triangular prisms laid along different cube axes, and $\arccos(7/9)$ for octahedra in a cube.

9 Symmetry (P7)

The configuration symmetry group of a record is the set of container symmetries, reflections included, that map the set of placed pieces onto itself. It is computed exactly on placed vertex sets at tolerance 10^{-4} edges. Of the 8,514 3D records, 7,738 are asymmetric and 776 have a nontrivial group (Figure 4a). The fully symmetric records (orders 24, 48 and 120) are all windows or shells. Symmetric seeding (134 records) was one of the search moves, but most symmetric records come from lattice cuts and contributed constructions. Relaxation from a symmetric start rarely keeps the symmetry.

10 Algebraic scales (P8)

A rigid record is a solution of its contact system: equalities between support functions of touching pieces and walls. Its scale is then algebraic. 1,624 records carry a conjectured closed form matched to 10^{-11}: 895 rational, 266 in $\mathbb{Q}(\sqrt{2})$, 185 in $\mathbb{Q}(\sqrt{3})$ and 148 in $\mathbb{Q}(\sqrt{5})$, the rest in nested or biquadratic fields. The field reflects the geometry of the contacts. Golden scales come from dodecahedral and icosahedral walls, silver ratios $\delta = 1 + \sqrt{2}$ from square antiprisms (PingYou's $S_2 \leq (5 - \delta)/\sqrt{\delta}$, $S_9 \leq 1 + 2/(11 - 4\delta)$, $S_{20} \leq 3 + 1/(\delta + 2)$ [8]), and $\sqrt{3}$ from hexagonal and dodecagonal prisms. A closed form is an exact statement only once the contact system is solved and certified; PingYou's "six-equation", "77-equation" and "115-equation" contact constructions are such certificates [8].

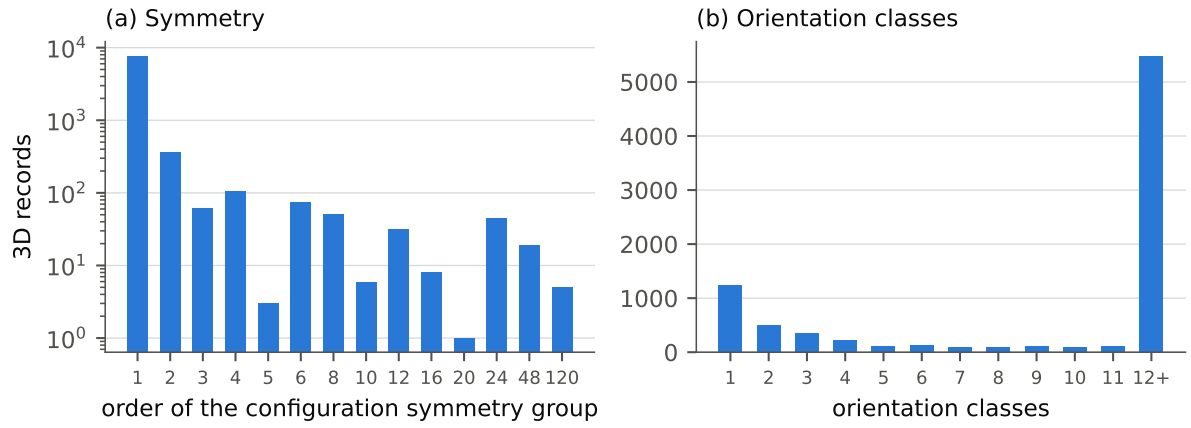


Figure 4: (a) Orders of configuration symmetry groups of the 8,514 3D records (log scale). (b) Number of orientation classes.

11 Extremal patterns: proven optimality (P9)

At the extreme of the classification are records whose scale is matched by a lower bound. Three mechanisms occur.

- **Volume tilings.** k^d cubes and k^3 triangular prisms at scale k (`Cubes.lean`, `Prisms.lean`).
- **Centre lemmas for $n = 2$ and plateaus.** If every piece at scale below s contains the container's centre, then $s(2) \geq s$. Proven instances include the cube centre lemma in dimensions 3 and 4 (plateaus $s = 2$ for $2 \leq n \leq 8$ cubes and $2 \leq n \leq 16$ tesseracts), and the octahedron lemma (2 octahedra in a cube, $\sqrt{2}$).
- **Elementary bounds.** The facet-width and two-ball lemmas prove optimality of 2 tetrahedra in a cube, octahedron, dodecahedron and icosahedron; 2 five-simplices in a 5-cube (the Paley construction); 2 five-cells in a 24-cell; and 2 16-, 24-, 120- and 600-cells in a tesseract or 16-cell.

These results are recorded in `docs/OPTIMALITY.md`. They also explain several plateau starts: the plateau begins where the lower bound is tight.

12 PingYou's sixteen constructions in the classification

Table 4 places the sixteen constructions of PingYou's notes [7] in the classification. All are in the catalogue. Four have since been improved by `konoha-ily2`, so they are historical in the sense of their notes. Five are now proved in Lean at the exact scale, and the others are certified records.

13 Classification by family

Table 5 classifies the records by family group. Prisms and antiprisms, and the Platonic pairs, are dominated by plateaus and lattice windows: their pieces tile or nearly tile. The Catalan and Archimedean families are mostly irregular, with fewer windows and fewer frozen subsets. Their records are individually tightened contact configurations, where arrows are the natural tool. In dimensions 4–6, plateaus and closed forms are common, coming from grids, composites and the $n = 2$ centre lemmas. Figure 5 shows how the records were produced.

pieces, container	count	exact scale	class	catalogue now	formal status
truncated cubes	$61 = (4^3 - 8) + 5$	$13 - 6\sqrt{2}$	P3 completion of P5	PingYou Ltd	certified
cubes, icosahedron	$32 = 8 + 6 \cdot 4$	$12 - 4\sqrt{5}$	P6 shell	konoha-ily2 (3.0291)	certified
cubes, dodecahedron	$40 = 14 + 14 + 6 + 6$	$5\varphi - 6$	P6 layers	PingYou Ltd	certified
cubes, dodecahedron	$29 = 9 + 11 + 9$	$5/\varphi^2$	P6, P8	konoha-ily2 (1.8595)	certified
cubes, dodecahedron	$30 = 9 + 12 + 9$	$23 - 13\varphi$	P6, P8	konoha-ily2 (1.9022)	certified
triakis tetrahedra	$16 = 4^2$	$(5/3)^2$	P4 recursion	PingYou Ltd	Lean (exact)
triakis tetrahedra	$64 = 4^3$	$(5/3)^3$	P4 recursion	PingYou Ltd	Lean (exact)
triangular prisms	$64 = 4^3$	4	P5 tiling, P9	PingYou Ltd	Lean (optimal)
truncated tetrahedra	$71 = 40 + 31$	13/3	P5 two orientations	PingYou Ltd	certified
truncated tetrahedra	$35 = 19 + 16$	11/3	P5 two orientations	PingYou Ltd	certified
truncated cubes	$56 = 4^3 - 2^3$	4	P5 Cartesian shell	PingYou Ltd	Lean (exact)
rhombic dodecahedra	$19 = 1 + 12 + 6$	3	P5 FCC shell	PingYou Ltd	Lean (exact)
rhombic dodecahedra	$44 = 2 + 8 + 12 + 12 + 8 + 2$	4	P5 FCC window	PingYou Ltd	Lean (exact)
height-2 triangular prisms, cube	$21 = 6 + 15$	3	P6 mixed layers	PingYou Ltd	certified
octagonal prisms, cube	$44 = 6 + 6 + 32$	$4 + 2\sqrt{2}$	P6, P8	PingYou Ltd	certified
height-3/2 dodecagonal prisms, cube	$20 = 2 \cdot 2 \cdot 5$	15/2	P5 column array	PingYou Ltd	certified

Table 4: PingYou's sixteen constructions [7]: number pattern, exact scale, class, current catalogue holder (with the current scale where it changed), and formal status.

Group	families	records	on a plateau	closed form	symmetric	lattice window	one orientation	frozen subsets
Platonic pairs	25	2,475	791	272	274	347	441	327
Archimedean	13	1,287	227	120	108	124	212	132
Catalan	13	1,287	99	72	110	42	215	50
Prisms and antiprisms	35	3,465	1,892	742	284	376	373	875
4D regular polytopes	36	1,404	297	300	–	–	–	–
5D regular polytopes	9	279	49	64	–	–	–	–
6D regular polytopes	9	207	47	54	–	–	–	–

Table 5: Pattern prevalence by family group: records on a plateau, with a closed form, with nontrivial symmetry, that are lattice windows, with a single orientation class, and frozen-subset pairs $f(k, N) = 1$ ($N - k \leq 6$). Geometric features are computed for 3D only.

14 Formalization

All theorems cited above are in `lean-optimality/` (Lean 4.31, Mathlib). The continuous-integration audit checks that every one depends only on the axioms `propext`, `Classical.choice` and `Quot.sound`.

Module	Theorems	Content
`Basic.lean`	`Packing.restrict`	deletion
`Patterns.lean`	`Packing.mono`, `compose`, `selfCompose`, `pow`	scale monotonicity, composition
	`of_separators`, `smulBoth`	separator certificates (F), rescaling
	`voronoi_disjoint`, `voronoi_packing`, `lattice_window`	Voronoi windows
`Windows.lean`	`rdo_window`, `tcu_window`, `rdo19/44/93`, `tcu56`	exact FCC and cubic windows
`Triakis.lean`	`ktt4`, `triakis_recursion`, `ktt16`, `ktt64`	the recursion 4^k at $(5/3)^k$

The formal results are of two kinds. The mechanisms are proved for arbitrary bodies and containers: deletion, monotonicity, composition, separator certificates and Voronoi windows. The exact instances are specific packings at exact algebraic scales; the catalogue certifies their padded counterparts by exact arithmetic. Descriptive classes (layers, orientation classes, symmetry) are measurements, not theorems.

15 Limitations

The classification is empirical: a record can belong to several classes, and the thresholds (tolerances, the cap of four orientation classes for windows, the gap $N - k \leq 6$ for inheritance) are

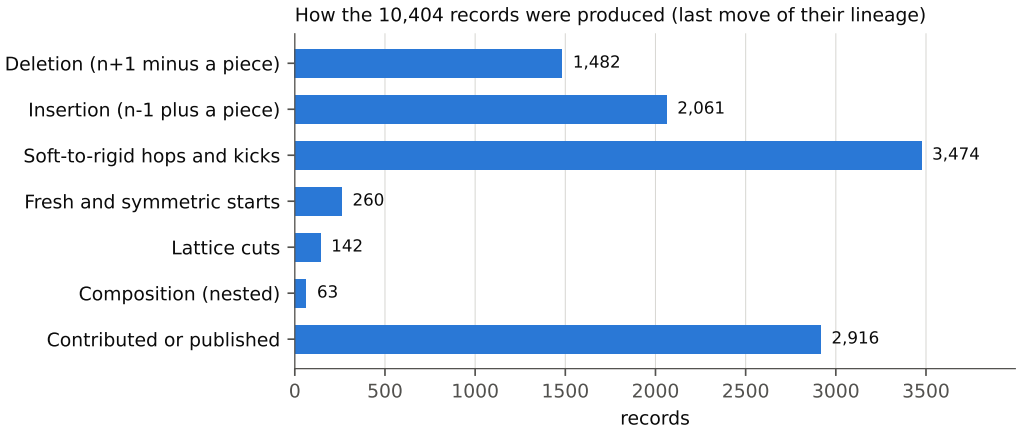


Figure 5: The last move in each record's lineage. Deletion and insertion together produced a third of the catalogue.

choices. The arrows reported here reproduce values PingYou Ltd announced first; they produced no bound below theirs, and their own witnesses await submission. Closed forms are conjectures until their contact systems are solved exactly. The formal layer proves the mechanisms and the listed instances, not the optimality of the windows. Only the records in Section 11 are proven optimal.

Acknowledgements

The arrows of Section 4, their F and P steps, and many of the number patterns of Sections 6–7 are due to PingYou Ltd [7, 8]. Their constructions also include the corner completion, golden shells and layers, finite Welsh windows, FCC shells and windows, the Cartesian shell, and mixed, quadratic-field and column layers. The catalogue combines searches and contributions by PingYou Ltd, `konoha-ily2` (Hirotaka Nakano), PolyPack (Michael Sinko), Berthold, Kamp, Mexi, Pokutta and Pólik [5], Hiroaki Hosono [6], Stian, Ryan Xu and others. Soft-to-rigid continuation is due to Yohei Nakajima [4]. AI coding and drafting tools, including Claude (Anthropic), assisted implementation, analysis, formalization and manuscript preparation under human direction. The Lean build and the exact certificates, not the tools, are the evidence for the formal and certified claims.

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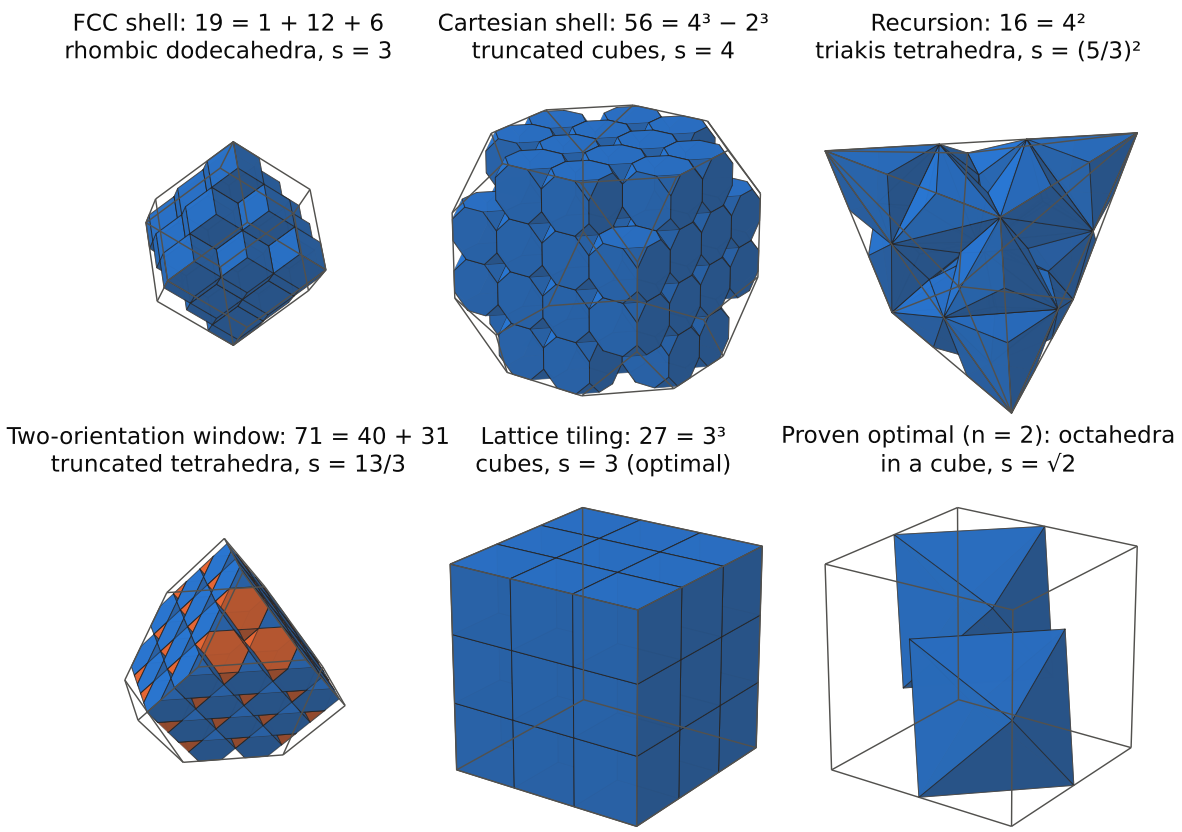


Figure 6: Pattern classes in records. Top: an FCC shell, a Cartesian shell and the triakis recursion (all proved in Lean at the exact scale). Bottom: a two-orientation window (orientation classes in blue and orange), a cube tiling, and a proven-optimal $n = 2$ record.

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